



Gabriel-Alin ANDREI

PhD Candidate in Industrial Engineering

National University of Science and Technology POLITEHNICA
Bucharest, Pitești University Centre

Mechanical Design Engineer

CER Cleaning Equipment, Curtea de Argeș, Romania

CONTACT INFORMATION

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EDUCATION

PhD Candidate in Industrial Engineering, 2026 to present

National University of Science and Technology POLITEHNICA Bucharest, Pitești University Centre

Topic: Research on the Mechanical Anchoring and In-Service Behavior of Metallic Inserts Integrated into Polymer Parts via FFF Additive Manufacturing.

Since: 21/09/2026

Master's studies in Engineering, 2024 to 2026

University of Pitești, Faculty of Mechanics and Technology

Programme: Product Manufacturing Engineering and Management

From: 05/10/2024 to 22/07/2026

Certification: Completed the Psycho-Pedagogical Training Programme (Level II)

Engineer, 2020 to 2024

University of Pitești, Faculty of Mechanics and Technology

Field: Mechanical Engineering Technology

From: 03/09/2020 to 28/07/2024

Certification: Completed the Psycho-Pedagogical Training Programme (Level I)

Natural Sciences, 2016 to 2020

Ion C. Brătianu National College, Pitești

From: 17/08/2016 to 20/06/2020

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer (01/08/2024 to present)

CER CLEANING EQUIPMENT, Curtea de Argeș, Romania

Mechanical Design Engineer - R&D Scrubbers, responsible for developing 3D concepts, designing mechanical components, and updating technical drawings for industrial scrubber systems. Inspect and validate supplier components according to technical specifications and tolerances, identify mechanical or fit-up issues, and propose effective corrective solutions.

Trainee (27/06/2023 to 01/10/2023)

CER CLEANING EQUIPMENT, Curtea de Argeș, Romania

Executed advanced 3D CAD modeling of plastic components, spline surfaces, and electrical harnesses in CATIA (Part Design, GSD). Validated the VD3 vacuum cleaner architecture via Digital Mock-Up (DMU), conducting clash detection, tolerance stack-up analysis, and assembly sequence simulations for manufacturability and ergonomics.

Trainee (18/9/2022 to 18/03/2023)

LEONI Wiring Systems S.R.L., Pitesti, Romania

Executed electrical functionality testing and quality verification across wire harness workstations, while leveraging Microsoft Excel for production data logging and analysis, optimizing workstation layouts for ergonomics and Lean efficiency, and implementing labeling protocols to ensure full product traceability.

RESEARCH INTERESTS

Additive Manufacturing and 3D Printing Technologies
Dimensional Accuracy & Thermal Behavior of 3D Printed Assemblies
Design for Additive Manufacturing (DfAM) & Product Development
Quality Assurance & Process Optimization in FDM/FFF Printing

SELECTED RESEARCH PROJECTS

Doctoral Research in Industrial Engineering | *POLITEHNICA Bucharest – Pitești University Centre* (2026 – Present)

Focus: Research on the Mechanical Anchoring and In-Service Behavior of Metallic Inserts Integrated into Polymer Parts via FFF Additive Manufacturing.

R&D Industrial Engineering Projects | *CER Cleaning Equipment (Kärcher Group)* (2024 – Present)

Focus: Development, 3D CAD design, and component validation for industrial scrubber systems, including supplier part verification and manufacturing process optimization.

SCIENTIFIC OUTPUT

2021 – 2026 – Student Scientific Communications Sessions, University of Pitești / POLITEHNICA Bucharest – Pitești University Centre

Participated in several student scientific communication sessions, presenting technical papers in the fields of mechanical engineering, manufacturing technologies, materials and 3D printing.

2023 – Computer-Aided Design Competition – Product Design, University of Pitești

Participant in the *Computer-Aided Design – Product Design* competition.

2022 – “C.C. Teodorescu” Professional-Scientific Competition – 3rd Prize

Awarded 3rd Prize for results in *Strength of Materials / Mechanical Engineering*.

AWARDS AND DISTINCTIONS

Valedictorian – Ranked 1st in the Bachelor’s graduating class, 2024.

TECHNICAL SKILLS

Software: CATIA V5, AutoCAD, SAP, AHP, Abaqus, MATLAB, Z-Suite, SinuTrain, Microsoft Office (Word, PowerPoint, Excel).

LANGUAGES

Romanian: Mother tongue

English: B1

French: A2

PERSONAL COMPETENCIES

Analytical thinking, adaptability and attention to detail

Strong communication and teamwork skills

Well-organized, proactive and solution-oriented

Committed to continuous learning and professional development

DATE: 2026

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